LAB REPORT

Title

Smart Traffic Management System Using C++

Objective

To develop a smart traffic management system that:

- Controls traffic automatically
- Detects traffic density
- Gives priority to ambulances
- Reduces congestion using intelligent decision-making

Introduction

Traditional traffic systems use fixed timing and cannot understand actual traffic conditions. This project simulates a smart traffic system that analyzes vehicle density and controls traffic flow automatically.

The system also provides emergency priority to ambulances for efficient transportation.

Features

- Four-lane traffic control
- Density-based signal management
- Ambulance priority system
- Smart lane selection
- Automatic signal switching
- Real-time traffic monitoring

Libraries Used

Library	Purpose
iostream	Input and output
thread	Multithreading
chrono	Time delay
mutex	Prevents data conflict
vector	Stores lanes
string	Stores names
cstdlib	Random values
ctime	Random seed

Main Modules

1. Detection System

Responsible Function:

`updateDensity()`

Simulates traffic sensors and updates vehicle density.

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2. Ambulance Detection System

`triggerAmbulance()`

Detects emergency vehicles and activates priority mode.

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3. Smart Decision System

`followSmartCycle()`

Finds the busiest lane and gives GREEN signal.

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4. Signal Control System

`setGreenLane()`

Controls which lane becomes GREEN.

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5. Display System

`displayAllLanes()`

Displays traffic status of all lanes.

Formula / Logic Used

The system selects the lane with maximum traffic density.

Logic:

\text{Lane Priority} = \max(\text{Vehicle Density})

Algorithm

1. Start system
2. Generate traffic density
3. Detect ambulance
4. Find busiest lane
5. Give GREEN signal
6. Display traffic status
7. Repeat continuously

Advantages

- Reduces congestion
- Saves fuel and time
- Supports emergency vehicles
- Improves traffic efficiency
- Demonstrates smart automation

Limitations

- Uses random density values
- No real sensors
- No graphical interface
- Single intersection only

Future Improvements

- AI camera integration
- IoT sensors
- Cloud monitoring
- Machine learning prediction
- Smart city implementation

Conclusion

The Smart Traffic Management System successfully demonstrates intelligent traffic control using C++. The project uses density-based decision-making and ambulance priority handling to simulate a modern smart traffic intersection.

Viva Questions

Q1: What is the purpose of this project?

To reduce congestion and automate traffic management.

Q2: Which function handles smart traffic decisions?

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followSmartCycle()
```

Q3: Why is mutex used?

To prevent simultaneous data access conflicts.

Q4: Which module handles ambulances?

`handleAmbulancePriority()`

Q5: What is the role of vehicle density?

It helps the system identify the busiest lane.

Q6: Which programming concepts are used?

- Classes
- Objects
- Loops
- Functions
- Multithreading
- OOP concepts